HOW TO DEPLOY ILLUMIO CORE FOR KUBERNETES WITH HELM CHART

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SUMMARY

How to deploy Illumio kubelink / c-ven using Helm Chart

OBJECTIVE

Steps to deploy Illumio kubelink / c-ven using Helm Chart

PROCEDURE

Starting with the Illumio Core for Kubernetes 3.0.0 release and later, the product (including C-VEN and Kubelink) is now deployed by using a Helm Chart. The product components and the Helm Chart are downloaded from a public container repository https://quay.io/repository/illumio/illumio

Reference: https://docs.illumio.com/core/22.5/Content/Guides/kubernetes-and-openshift/deployment-helm/_ch-deployment-helm.htm

Below are steps on how to deploy Illumio kubelink / c-ven using Helm Chart:

How to deploy Illumio kubelink / c-ven using Helm Chart:

1.
Log into PCE UI and create Container Cluster under '`Menu > infrastructure > container cluster`'. Make note of the cluster id and cluster token as it will be needed to complete a yaml file at a later point. This information will only be shown once in the PCE UI.

```
Cluster id: 92dd0ed4-d0b5-4136-81cc-54cd07772245
Cluster token: 1_a06e203e693ab8b3c58b159858be411afc4662decada07f798a763c78b9bed53
```

2.
After creation of container cluster go to '`Menu > Infrastructure > Container Clusters`', select the newly created container cluster. Go to 'Container Workload Profile > Edit default settings', in this example we will add an environment label 'minikube1_env_container_annotations', then save. It is recommended to add a default label which will be shared by all the Container Workload Profiles. All nodes in the kubernetes cluster will have this label by default for this specific container cluster. All container workload will also have this same container annotations labels.

3.
Create a pairing profile to pair CVEN to PCE. Go to '`Menu > Workloads and VENs > Pairing Profile > Add`'. Illumio recommends all pairing profiles for Kubernetes nodes to not use Enforced policy state. Use Build or Test mode for initial configuration.

4.
Install Helm on kubernets cluster.

```
$ curl -fsSL -o get_helm.sh https://raw.githubusercontent.com/helm/helm/main/scripts/get-helm-3
$ chmod 700 get_helm.sh
$ $ ./get_helm.sh
```

reference: https://helm.sh/docs/intro/install/

5.
Initialize helm chart repository and update it:

```
$ helm repo add bitnami https://charts.bitnami.com/bitnami
$ helm repo update
```

reference: https://helm.sh/docs/intro/quickstart/

6.
Prepare an illumio-values.yaml file with the following mandatory parameter values in the kubernetes cluster node. Replace pce_uri with your PCE FQDN, cluster_id and cluster_token from step 1 above after creating container cluster. During initial Helm install, the keys in this file will be used for depolying both kubelink and C-VEN.

Explanation of each key:

pce_url: URL_PORT # PCE URL with port, e.g. mypce.example.com:8443
cluster_id: ILO_CLUSTER_UUID # Cluster ID from PCE, e.g. cc4997c1-40...
cluster_token: ILO_CLUSTER_TOKEN # Cluster Token from PCE, e.g. 1_170b...
cluster_code: ILO_CODE # Pairing Profile key from PCE, e.g. 1391c...
containerRuntime: containerd # Container runtime engine used in cluster, allowed values are [containerd, docker, crio]
containerManager: kubernetes # Container manager used in cluster, allowed values are [kubernetes, openshift]

If you are using a private PKI to sign the PCE certificate, make sure you add to ```illumio-values.yaml``` the references to the root CA certificate that signed the PCE certificate, as shown below. Used to mount directories to Kubelink and C-VEN containing certificates:

```
extraVolumeMounts:
  - name: root-ca
    mountPath: /etc/pki/tls/ilo_certs/
    readOnly: false
extraVolumes:
  - name: root-ca
    configMap:
      name: root-ca-config
```

Option to ignore certificate verification of PCE, Skip certificate verification of `pce_url` in Kubelink:

```
ignoreCert: "false"
```

The verbosity level of information sent to PCE, ["0", "1", "2"] default is '0':

```
verbosity: "0"
```

Example of illumio-values.yaml :

```
pce_url: pce.domain.com:8443
cluster_id: 92dd0ed4-d0b5-4136-81cc-54cd07772245
cluster_token: 1_a06e203e693ab8b3c58b159858be411afc4662decada07f798a763c78b9bed53
cluster_code:
15cc2fca4319ce5edafa189bc94ff217703e2af0fc879d1cb6bd3de5b3ae4a9f68d67ec7aab0e69bd
containerRuntime: docker
containerManager: kubernetes
```

```
    extraVolumeMounts:
      - name: root-ca
        mountPath: /etc/pki/tls/ilo_certs/
        readOnly: false
    extraVolumes:
      - name: root-ca
        configMap:
          name: khoa-root-ca-config
    ignoreCert: "false"
    verbosity: 0
```

7.
Note: This step is optional for environment where a private PKI was used to sign PCE certificate.

Create the configmap for named "khoa-root-ca-config" prior to using Helm to install since it will be used for the self-signed CA certificate of the PCE. The namespace "illumio-system" is the default namespace used for kubelink/C-VEN deployment in a kubernetes cluster. The file "./private_certificate/DigiCertCA.crt" is the private CA certificate.

```
$ kubectl create namespace illumio-system
$ kubectl -n illumio-system create configmap khoa-root-ca-config --from-
file=./private_certificate/DigiCertCA.crt
```

Done.

8.
Add hostname resolution of PCE before installing Helm chart since it might not be resolvable. If PCE hostname is resolvable in cluster environment, skip this step.

There should be a configmap "coredns" in a kubernetes environment. Edit and add the PCE hostname resolution:

`$ kubectl -n kube-system get configmap` [get list of configmap]

```
kubectl -n kube-system get configmap
NAME                                 DATA   AGE
coredns                              1      6d19h
extension-apiserver-authentication   6      6d19h
kube-proxy                           2      6d19h
kube-root-ca.crt                     1      6d19h
kubeadm-config                       1      6d19h
kubelet-config                       1      6d19h
```

`$ kubectl -n kube-system edit configmap coredns` [edit configmap coredns under data.Corefile.:53.hosts]

```
hosts {
   192.168.49.1 host.minikube.internal
   10.2.177.204 pce.privatedomain.com
   fallthrough
}
```

`$ kubectl -n kube-system rollout restart deployment coredns` [restart coredns deployment for changes to take effect]

9.
Use dnsutils to check with dig/nslookup and validate name resolution works cluster wide.
Once the pod is up and running on the kubernetes 'default' namespace use dig/nslookup to test::

```
$ kubectl apply -f https://k8s.io/examples/admin/dns/dnsutils.yaml
$ kubectl get pods dnsutils
$ kubectl exec dnsutils -ti -- dig pce.privatedomain.com
```

```
kubectl exec dnsutils -ti -- dig pce.privatedomain.com

; <<>> DiG 9.9.5-9+deb8u19-Debian <<>> pce.privatedomain.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 60457
;; flags: qr aa rd; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1
;; WARNING: recursion requested but not available

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 4096
;; QUESTION SECTION:
;pce.privatedomain.com. IN      A

;; ANSWER SECTION:
pce.privatedomain.com. 30       IN A    10.2.177.204

;; Query time: 2 msec
;; SERVER: 10.96.0.10#53(10.96.0.10)
;; WHEN: Fri Feb 17 02:02:05 UTC 2023
;; MSG SIZE  rcvd: 115
```

Tail the coredns pod logs to validate a dns request was made from the dig command, output below indicates a successful dns query.

```
$ kubectl -n kube-system logs coredns-c477fbf6f-9ntxw -f --tail=5
```

```
kubectl -n kube-system logs coredns-c477fbf6f-9ntxw -f --tail=5

[INFO] 10.244.0.35:52622 - 38663 "A IN pce.privatedomain.com. udp 64 false 4096" NOERROR
qr,aa,rd 104 0.000489814s
```

10.
Note: If the namespace illumio-system was never created from prior step add the parameter "create-namespace" to the "helm install" command. Step below assume illumio-system namespace was already created.

Deploy the helm chart. The "--debug --dry-run" is good practice to check for errors before installing.

```
$ helm install --debug --dry-run illumio -f illumio-values.yaml oci://quay.io/illumio/illumio
--namespace illumio-system
$ helm install illumio -f illumio-values.yaml oci://quay.io/illumio/illumio --namespace
illumio-system
```

Other helm commands to use during install for troubelshooting:

`$ helm show all oci://quay.io/illumio/illumio` [explain keys used during helm install]

`$ helm uninstall illumio --namespace illumio-system` [uninstall kubelink/C-VEN , use to re-deploy after changes]

`$ kubectl delete namespace illumio-system` [after uninstall also delete the illumio-system namespace]

11.
After helm install of Illumio core for Kubenetes, verify both kubelink and C-VEN pods are in running state, use command below to check:

```
$ kubectl -n illumio-system get pods
```

```
kubectl -n illumio-system get pods
NAME                                READY   STATUS    RESTARTS   AGE
illumio-kubelink-67865896b8-wf2qs   1/1     Running   0          2d
illumio-ven-wdljt                   1/1     Running   0          2d
```

```
$ kubectl -n illumio-system describe pod illumio-kubelink-67865896b8-wf2qs
```

```
kubectl -n illumio-system describe pod illumio-kubelink-67865896b8-wf2qs
Name:             illumio-kubelink-67865896b8-wf2qs
Namespace:        illumio-system
Priority:         0
Service Account:  illumio-kubelink
Node:             minikube/192.168.49.2
Start Time:       Wed, 15 Feb 2023 15:50:35 -0500
Labels:           app=illumio-kubelink
                  pod-template-hash=67865896b8
Annotations:      <none>
Status:           Running
IP:               10.244.0.37
IPs:
  IP:           10.244.0.37
Controlled By:  ReplicaSet/illumio-kubelink-67865896b8
Containers:
  illumio-kubelink:
    Container ID:   docker://980b0d2151e9fd92e2d6790530f9899cf4584bb9125fa929a3b6d741f3f68796
    Image:          quay.io/illumio/kubelink:3.2.0.7b7a38
    Image ID:       docker-
pullable://quay.io/illumio/kubelink@sha256:059273dcb7fb1e84731445b1cffef5844d44c5a03e489821223585cdaf
b61225
    Port:           <none>
    Host Port:      <none>
    State:          Running
      Started:      Wed, 15 Feb 2023 15:50:37 -0500
    Ready:          True
    Restart Count:  0
    Liveness:       http-get http://:8081/healthz delay=30s timeout=1s period=30s #success=1
#failure=3
    Readiness:      http-get http://:8081/readyz delay=10s timeout=1s period=10s #success=1
#failure=5
    Environment:
      ILO_SERVER:         <set to the key 'ilo_server' in secret 'illumio-kubelink-config'>
Optional: false
      ILO_CLUSTER_UUID:   <set to the key 'ilo_cluster_uuid' in secret 'illumio-kubelink-config'>
Optional: false
      ILO_CLUSTER_TOKEN:  <set to the key 'ilo_cluster_token' in secret 'illumio-kubelink-config'>
Optional: false
      IGNORE_CERT:        <set to the key 'ignore_cert' in secret 'illumio-kubelink-config'>
Optional: true
```

```
      VERBOSITY:          <set to the key 'verbosity' in secret 'illumio-kubelink-config'>
Optional: true
    Mounts:
      /etc/pki/tls/ilo_certs/ from root-ca (rw)
      /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-5b89m (ro)
Conditions:
  Type              Status
  Initialized       True
  Ready             True
  ContainersReady   True
  PodScheduled      True
Volumes:
  root-ca:
    Type:      ConfigMap (a volume populated by a ConfigMap)
    Name:      khoa-root-ca-config
    Optional:  false
  kube-api-access-5b89m:
    Type:                    Projected (a volume that contains injected data from multiple sources)
    TokenExpirationSeconds:  3607
    ConfigMapName:           kube-root-ca.crt
    ConfigMapOptional:       <nil>
    DownwardAPI:             true
QoS Class:                   BestEffort
Node-Selectors:              <none>
Tolerations:                 node.kubernetes.io/not-ready:NoExecute op=Exists for 300s
                             node.kubernetes.io/unreachable:NoExecute op=Exists for 300s
Events:                      <none>
```

```
$ kubectl -n illumio-system describe pod illumio-ven-wdljt
```

```
kubectl -n illumio-system describe pod illumio-ven-wdljt
Name:             illumio-ven-wdljt
Namespace:        illumio-system
Priority:         0
Service Account:  illumio-ven
Node:             minikube/192.168.49.2
Start Time:       Wed, 15 Feb 2023 16:03:50 -0500
Labels:           controller-revision-hash=6844858597
                  name=illumio-ven
                  pod-template-generation=2
Annotations:      <none>
Status:           Running
IP:               192.168.49.2
IPs:
  IP:           192.168.49.2
Controlled By:  DaemonSet/illumio-ven
Containers:
  illumio-ven:
    Container ID:  docker://84485221ce333b8aa30d86ab0ee074017d1b95b9dddab1b887090e85403e22ae
    Image:         quay.io/illumio/cven:22.5.11-9729
    Image ID:      docker-
pullable://quay.io/illumio/cven@sha256:240c6413f9f5481ab10db6055c6a06da0437b49fb984c6c0504ad918345f56
cb
    Port:          <none>
    Host Port:     <none>
    Command:
      /ven-init
      activate
    State:          Running
      Started:      Wed, 15 Feb 2023 16:03:51 -0500
    Ready:          True
    Restart Count:  0
    Limits:
      memory:  300Mi
    Requests:
      memory:  100Mi
    Environment:
      ILO_SERVER:         <set to the key 'ilo_server' in secret 'illumio-ven-config'>  Optional:
false
      ILO_CODE:           <set to the key 'ilo_code' in secret 'illumio-ven-config'>    Optional:
false
```

```
      ILO_K8S_NODE_NAME:   (v1:spec.nodeName)
    Mounts:
      /etc/machine-id from machine-id (ro)
      /etc/pki/tls/ilo_certs/ from root-ca (rw)
      /lib/modules from modules (ro)
      /opt/illumio_ven_data from illumio-ven-data (rw)
      /var/run/docker.sock from unixsocks (rw)
      /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-8ld2g (ro)
Conditions:
  Type              Status
  Initialized       True
  Ready             True
  ContainersReady   True
  PodScheduled      True
Volumes:
  unixsocks:
    Type:          HostPath (bare host directory volume)
    Path:          /var/run/docker.sock
    HostPathType:  Socket
  illumio-ven-data:
    Type:          HostPath (bare host directory volume)
    Path:          /opt/illumio_ven_data
    HostPathType:  DirectoryOrCreate
  modules:
    Type:          HostPath (bare host directory volume)
    Path:          /lib/modules
    HostPathType:  Directory
  machine-id:
    Type:          HostPath (bare host directory volume)
    Path:          /etc/machine-id
    HostPathType:  File
  root-ca:
    Type:      ConfigMap (a volume populated by a ConfigMap)
    Name:      khoa-root-ca-config
    Optional:  false
  kube-api-access-8ld2g:
    Type:                    Projected (a volume that contains injected data from multiple sources)
    TokenExpirationSeconds:  3607
    ConfigMapName:           kube-root-ca.crt
    ConfigMapOptional:       <nil>
    DownwardAPI:             true
QoS Class:                   Burstable
```

```
Node-Selectors:              <none>
Tolerations:                 node-role.kubernetes.io/control-plane:NoSchedule
                             node-role.kubernetes.io/master:NoSchedule
                             node.kubernetes.io/disk-pressure:NoSchedule op=Exists
                             node.kubernetes.io/memory-pressure:NoSchedule op=Exists
                             node.kubernetes.io/network-unavailable:NoSchedule op=Exists
                             node.kubernetes.io/not-ready:NoExecute op=Exists
                             node.kubernetes.io/pid-pressure:NoSchedule op=Exists
                             node.kubernetes.io/unreachable:NoExecute op=Exists
                             node.kubernetes.io/unschedulable:NoSchedule op=Exists
Events:                      <none>
```

12.
After a successful helm chart install, the PCE will log the events under '`Menu > Troubleshooting > Events`' below:
Successful kubelink install will log container_cluster_update and container_cluster.update_services:

Customize columns | 50 per page | 1 – 50 of 118 Total

Event	Description	Severity	Status	Timestamp	Generated By
container_cluster.update_services	Container Workload Update Services	Informational	Success	02/14/2023, 14:28:32	[illegible]
container_cluster.update	Container cluster updated	Informational	Success	02/14/2023, 14:28:16	[illegible]
container_cluster.update	Container cluster updated	Informational	Success	02/14/2023, 14:28:15	[illegible]
container_cluster.update	Container cluster updated	Informational	Success	02/14/2023, 14:28:15	[illegible]
user.login	User login	Informational	Success	02/14/2023, 14:25:54	[illegible]

Successful C-VEN install will log agent.activate, agent.machine_identifier, workload_interfaces.update, agent.update_running_containers, and agent.update events:

Customize columns | 50 per page | 1 – 50 of 172 Total

Event	Description	Severity	Status	Timestamp	Generated By
user.login	User login	Informational	Success	02/15/2023, 13:13:45	[illegible]
user.sign_in	User session created	Informational	Success	02/15/2023, 13:13:41	[illegible]
user.sign_in	User session created	Informational	Success	02/15/2023, 13:13:36	[illegible]
agent.update	VEN properties updated	Informational	Success	02/15/2023, 13:06:05	[illegible]
agent.update	VEN properties updated	Error	Failure	02/15/2023, 13:04:06	[illegible]
agent.goodbye	Workload shutdown	Informational	Success	02/15/2023, 13:03:49	[illegible]
agent.update	VEN properties updated	Informational	Success	02/15/2023, 13:03:38	[illegible]
agent.update_running_containers	VEN updated running containers	Informational	Success	02/15/2023, 13:03:37	[illegible]
workload_interfaces.update	Workload interfaces updated	Informational	Success	02/15/2023, 13:03:27	[illegible]
agent.machine_identifier	VEN machine identifiers updated	Informational	Success	02/15/2023, 13:03:25	[illegible]
agent.activate	VEN paired	Informational	Success	02/15/2023, 13:03:23	[illegible]
user.sign_out	User session terminated	Informational	Success	02/15/2023, 12:55:50	[illegible]

Done.

13.
Items to check if C-VEN fails to activate or maintain a running state.

If the C-VEN fail to start with the error below after using the 'describe' command, it indicate the configmap was not created or missing since the illumio-values.yaml files have an extraVolumes set, check Step 7 again:

```
kubectl -n illumio-system describe pod  illumio-ven-j2wqj

Events:
  Type     Reason       Age                  From               Message
  ----     ------       ----                 ----               -------
  Normal   Scheduled    12m                  default-scheduler  Successfully assigned illumio-
system/illumio-ven-j2wqj to minikube
  Warning  FailedMount  11m (x6 over 12m)    kubelet            MountVolume.SetUp failed for volume
"root-ca" : configmap "khoa-root-ca-config" not found
  Normal   Pulled       10m (x4 over 11m)    kubelet            Container image
"quay.io/illumio/cven:22.5.11-9729" already present on machine
  Normal   Created      10m (x4 over 11m)    kubelet            Created container illumio-ven
  Normal   Started      10m (x4 over 11m)    kubelet            Started container illumio-ven
  Warning  BackOff      2m5s (x42 over 11m)  kubelet            Back-off restarting failed container
illumio-ven in pod illumio-ven-j2wqj_illumio-system(758cfcec-b8e6-4afb-abe7-1c0aa183e70a)
```

If the C-VEN pod have a status of 'CrashLoopBackOff' during the initial install, it may indicate the PCE is unreachable/unresolvable.

If the C-VEN fails activation with the error below in the logs, it may indicate PCE is unreachable/unresolvable:

```
kubectl -n illumio-system logs illumio-ven-j2wqj --tail=5
Hostname is minikube
Pairing with PCE pce.privatedomain.com on port 8443
Checking Runtime Environment..........
Illumio VEN pair currently in progress. Please retry later.
ERROR: activation failure: 1
```

If there is terminal access or tail of the agentmgr.logs you may see the error below indicating network issues reaching the PCE due to hostname resolution. Use the command to gain terminal access to a pod:

```
$ kubectl -n illumio-system exec illumio-ven-wdljt -ti -- tail
/opt/illumio_ven_data/log/agentmgr.log
```

```
C-VEN agentmgr.log:

$ kubectl -n illumio-system exec illumio-ven-wdljt -ti -- tail /opt/illumio_ven_data/log/agentmgr.log

2023-02-15T04:40:41.066Z 1384075 INFO:: Using initial MasterConfig server pce.privatedomain.com:8443
2023-02-15T04:40:41.271Z 1384075 ERROR:: Not all response headers are received.
2023-02-15T04:40:41.271Z 1384075 WARNING:: ConnMgr: DIRECT connection failed.
2023-02-15T04:40:41.271Z 1384075 ERROR:: [ApiHelper] Failed to GET
https://pce.privatedomain.com:8443/api/v25/orgs/0/agents/0/master_config (internal error 56)
```

If the C-VEN have trouble reaching the PCE due to hostname resolution, one option is to add the 'hostAliases' to illumio-ven daemonset since the C-VEN may be on a different network plane than the coreDNS pod's network plane. The 'hostAliases' is similar to editing '/etc/hosts' file on a linux system.
A DaemonSet ensures that all (or some) Nodes run a copy of a Pod. As nodes are added to the cluster, Pods are added to them. As nodes are removed from the cluster, those Pods are garbage collected. Deleting a DaemonSet will clean up the Pods it created. (reference: Daemonset)

Find the C-VEN daemonset:

```
kubectl -n illumio-system get daemonset
NAME          DESIRED   CURRENT   READY   UP-TO-DATE   AVAILABLE   NODE SELECTOR   AGE
illumio-ven   1         1         1       1            1           <none>          2d2h
```

Edit the daemonset and add the hostAliases under (spec.template.spec.hostAliases), then save changes and delete current C-VEN pod so the new pod will take in the changes.

```
kubectl -n illumio-system edit daemonset illumio-ven

hostAliases:
      - hostnames:
        - pce.privatedomain.com
        ip: 10.2.177.204
```

14.
If kubelink fails to connect to the PCE the errors below maybe seen from the logs. Check the coredns pods and validate PCE hostname is resolvable, review step 8, troubleshoot with step 9:

```
$ kubectl -n illumio-system logs illumio-kubelink-cb99f47f6-qp95g
```

```
kubectl -n illumio-system logs illumio-kubelink-cb99f47f6-qp95g

2023-02-14T22:17:29.260Z  INFO  Possible verbosity levels: {0: Info(default), 1: Debug, 2: K8s Api
responses}, verbosity is controlled with the "VERBOSITY" env. variable
2023-02-14T22:17:29.261Z  INFO  Current "VERBOSITY" {"verbosity": 0}
2023-02-14T22:17:29.310Z  INFO  controller-runtime.metrics  Metrics server is starting to listen
{"addr": ":8080"}
2023-02-14T22:17:29.310Z  INFO  pkg.pcelink Searching for custom certificates {"directory":
"/etc/pki/tls/ilo_certs"}
2023-02-14T22:17:29.337Z  INFO  pkg.pcelink Loading custom certificates from files  {"files":
["/etc/pki/tls/ilo_certs/..2023_02_14_22_14_41.829760416/DigiCertCA.crt",
"/etc/pki/tls/ilo_certs/DigiCertCA.crt"]}
2023-02-14T22:17:29.337Z  INFO  pkg.pcelink Reading certificate file  {"file":
"/etc/pki/tls/ilo_certs/..2023_02_14_22_14_41.829760416/DigiCertCA.crt"}
2023-02-14T22:17:29.338Z  INFO  pkg.pcelink Reading certificate file  {"file":
"/etc/pki/tls/ilo_certs/DigiCertCA.crt"}
2023-02-14T22:17:29.338Z  INFO  pkg.pcelink Loaded custom certs to CertPool
2023-02-14T22:17:29.338Z  INFO  pkg.pcelink Initialized PCE link instance
2023-02-14T22:17:29.339Z  INFO  pkg.pcelink Sending message to PCE  {"updateKind": "heartbeat",
"url": "https://pce.privatedomain.com:8443/api/v2/orgs/ILO/container_clusters/92dd0ed4-d0b5-4136-
81cc-54cd07772245/put_from_cluster"}
2023-02-14T22:17:29.359Z  INFO  pkg.pcelink PCE response  {"request_id": "", "status_code": 404,
"responseBody": "<!DOCTYPE html><html><head><title></title><meta charset=\"utf-8\"><meta
name=\"viewport\" content=\"width=device-width,initial-scale=1\"><link rel=\"shortcut icon\"
href=\"favicon.ico\"><link rel=\"stylesheet\" href=\"/static/css/web.css\"></head><body class=\"il-
static-page\"><header><div><img src=\"/static/images/alert_icon.svg\" height=\"14px\" width=\"14px\">
</div></header><section><div class=\"il-logo-border\"><img width=\"154\" height=\"40\"
src=\"/static/images/logo.svg\" alt=\"Illumio\"></div><h1></h1><div>If you reached this page, please
let us know at <a href=\"mailto:support@illumio.com\">support@illumio.com</a> so we can fix it.
</div></section><footer>&copy; 2013-2023 Illumio, Inc. All Rights Reserved.</footer></body></html>"}
2023-02-14T22:17:29.359Z  ERROR pkg.pcelink Received a non retriable error  {"status_code": 404,
"error": "non retriable error"}
```

Reference:
Deploy with Helm Chart https://docs.illumio.com/core/22.5/Content/Guides/kubernetes-and-openshift/deployment-helm/deploy-with-helm-chart.htm
Helm Docs: https://helm.sh/docs/chart_template_guide/getting_started/

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